Lecture plan · for the lecturer

Designing a data-science module

Data science · University · Lecture 4

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Align outcomes and assessment.
- Design a staged, real project.
- Write a rubric that teaches.

Activity: Align a module

1. Warm-up (5 min): Ask colleagues to sort last year's exam questions into 'calculate' and 'judge'.
2. Teach (10 min): Show an alignment table and the six GAISE recommendations.
3. Map (20 min): Pairs map their own module's outcomes to its assessments and mark the gaps.
4. Draft (20 min): Write four criteria and four levels for a project, then swap and critique.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which outcome has no assessment?
- How will you check reproducibility at scale?
- What feedback will the draft get?

Exit ticket

- Write one outcome with an active verb and the task that assesses it.

Try it in your next lecture
Take your module's learning outcomes, underline each verb, and write next to it the assessment that measures it; any verb with nothing beside it needs a new task.

Seminar sheet · for students

Designing a data-science module

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Name _____ Class _____ Date _____

1 **When should students first see the project rubric?**

◯ When the project is set ◯ With their final grade ◯ Never, to avoid gaming it

2 **Complete the weighting.**

A module gives 40% to the project and 25% to labs. If the exam gets the rest, it is worth _____ %. If the project's report is half its weight, the report is worth _____ %.

Word bank: 35 · 25 · 40 · 20

3 **Which outcomes are written so they can be assessed? Choose all that apply.**

☐ Justify the choice of a model for a given dataset ☐ Understand regression

☐ Critique the causal claim in a published study ☐ Appreciate the role of data in society

4 **Requiring a reproducible notebook makes copying results from another group harder.**

◯ True ◯ False

5 **Match each piece of feedback to Hattie and Timperley's question it answers.**

"The goal is a defensible causal claim."	• • • •	Where to next?
"Your DAG omits prior grade."	• • • •	How am I going?
"Add prior grade and rerun the model."	• • • •	Where am I going?

✎ **Which assessment in your module rewards calculation where the outcome asks for judgement? What would you replace it with?**